

1. Raw data (Feng et al., 2019, NASDAQ)

1,026 tickers $\times$ 1,246 trading days;
industry relation file (97 industries)

2. Preprocessing

Treat the -1234 sentinel as missing
Features: 5-, 10-, 20-, 30-day moving averages + close ratio,
each normalized by the stock's maximum price

3. Split by date (same for all models)

train: days 0-755 | validation: 756-1007 | test: 1008-

4a. Static graph

Industry one-hot,
1,026 $\times$ 1,026;
constant during training

4b. Dynamic graph

Rolling 60-day
correlation, $|\text{corr}| > 0.5$;
windows built
within each split

5a. Baseline

Rank_LSTM
no relation graph

5b. RSR

baseline's exported LSTM
embedding + attention
over the static graph

5c. Dynamic relation

(extension): same as RSR,
graph fed per
60-day window

6. Training (identical for all three models)

Adam, seq = 4, unit = 32; loss = MSE + $\alpha \times$ pairwise ranking penalty
10-epoch pilot, then matched 50-epoch runs

7. Evaluation on the test split

MSE | mrrt (top-pick reciprocal rank) | btl (long-only backtest return)